

Uriel Castorela Cuevas

AI Engineering Student · Multi-Tenant Architectures · Fintech · Algorithmic Optimization

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PROFESSIONAL SUMMARY

Software engineer focused on building high-availability SaaS platforms and robust transactional engines. Experienced in designing secure multi-tenant architectures with strict data isolation and dynamic routing. Specialized in resolving concurrent-processing bottlenecks through asynchronous refactoring and algorithmic optimization.

FEATURED TECHNICAL PROJECTS

Castoralia — Multi-Tenant SaaS Platform for Clinical Management

- Ensured cryptographic isolation and safe data concurrency for multiple dental clinics, achieving 100% data-integrity retention through a multi-tenant architecture based on dynamic subdomain assignment and routing.
- Designed the system end to end, from branding conception to relational database normalization, supporting clinical records at scale.

FinWise — Financial Sustainability Transaction Engine

Capital One Hackathon

- Refactored a blocking bottleneck into an asynchronous architecture, enabling support for massive concurrent user spikes with no service degradation.
- Implemented a precise debt-management calculation engine, governed by a strict finite state machine (safe cycles $8 \rightarrow 9 \rightarrow 4 \rightarrow 0$) and dynamic algebraic-expression evaluation to compute amortizations while respecting operator precedence (e.g. $8 + 6 / 2 * (3 + 1)$).

Spatial-Resolution & AI Solvers — Algorithmic Optimization

- Reduced memory footprint by optimizing informed-search algorithms (GBFS and A*) in high-complexity environments, scaling from 9×7 grids to higher-dimensional problems on the native Python 3.14 runtime.

EDUCATION

Escuela Superior de Cómputo (ESCOM — IPN) — B.S. in Artificial Intelligence Engineering

Relevant coursework: Systems Architecture, Algorithms, Data Structures

CECyT 9 "Juan de Dios Bátiz" (IPN) — Technician in Programming

TECHNICAL SKILLS

Languages: Python | TypeScript | Java · **Backend/SaaS:** NestJS | React | Django · **Databases:** PostgreSQL | MySQL | Multi-Tenant · **Core:** OOP | A* | State Machines